Code No: 136AQ **R16**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech III Year II Semester (Special) Examinations, January/February - 2021
COMPILER DESIGN
(Common to CSE, IT)

Time: 2 Hours **Max. Marks: 75**

Answer any five questions
All questions carry equal marks
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1. Give the structure of a compiler and explain various phases of compilation. [15]

2. Compute Firsts and Follows and construct LL(1) parser table for the following grammar after eliminating ambiguity.
   $R \rightarrow R + R$, $R \rightarrow R * R$, $R \rightarrow R / R$, $R \rightarrow (R)$, $R \rightarrow id$. [15]

3.a) How are inherited attributes differ from synthesized attributes?
b) Construct the SDD to convert infix notation to postfix notation. [5+10]

4.a) What is a type checker? Give the specification of a simple type checker.
b) Write the quadruples, triples and indirect triples for the expression: [7+8]
   a * b – (c – d) + b * a – (c – d)

5. Explain various storage allocation strategies and also mention their merits and demerits. [15]

6.a) What is a Basic Block? How to construct a Basic block?
b) Explain Peephole optimization with an illustrative example. [7+8]

7. Explain the principal sources of code optimization with merits and demerits of each. [15]

8. Explain the following:
   a) Basic elements of Data-Flow Analysis.
   b) Register Allocation. [8+7]

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used paper ran-feb-2021